

Stephen Steinle

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[Website](#) | [Google Scholar](#) | [LinkedIn](#)

Research Focus

Conversational strategic reasoning, natural language processing, cognitive modeling, persona evaluation, legal retrieval and entailment, and syntax-guided code generation.

Education

Ph.D., Computer Science | University of South Florida

Expected May 2027

B.S., Computer Science | Dakota State University

May 2022

B.S., Information Technology | University of South Florida

December 2018

Research and Professional Experience

Graduate Research Assistant | AMHR Lab, University of South Florida

August 2022 - Present

Develop NLP and large language model methods for strategic dialogue, including stance, goal, and strategy inference from conversational context.

Design controlled reasoning evaluations to identify the effects of missing information, constraints, and prompting assumptions.

Software Development Intern | Amazon Ring

August - November 2025

Built repository and documentation search using LLM-expanded keyword retrieval and query-focused document summaries.

Implemented an MCP service connecting Cursor with OpenSearch queries and keyword-usage summaries.

Application Developer Intern | CACI Inc.

May 2022 - March 2024

Implemented anomaly-detection workflows over vehicle traffic data to support investigation and triage.

Machine Learning Intern | Air Force Research Laboratory

May - August 2023

Built a natural-language action legality system that maps utterances to structured actions and checks permissibility against a ruleset and game state.

Cryptologic Technician Networks | United States Navy

October 2014 - August 2021

Performed cyber intelligence and cybersecurity operations, including incident response and network auditing.

Publications

Curiosity Exploration Styles in Word Association Tasks

Stephen Steinle, My Duong, Suraj Kumar, Rima El Brouzi, John Licato

Proceedings of the 47th Annual Meeting of the Cognitive Science Society (CogSci) (2025)

A survey of fatigue measures and models

Antonio Laverghetta Jr., Minh Tran, Alec Braynen, **Stephen Steinle**, Bekhzodbek Moydinboyev, Heba Daas, John Licato

The Journal of Defense Modeling and Simulation, 22(2) (2025), pp. 147-173

First published online in March 2023; journal issue published in April 2025.

Do Persona-Infused LLMs Affect Performance in a Strategic Reasoning Game?

John Licato, **Stephen Steinle**, Brayden Hollis

IJCNLP-AACL 2025 (2025), pp. 3497-3528

AMHR COLIEE 2024 Entry: Legal Entailment and Retrieval

Animesh Nigohkar, Kenneth Jiang, Logan Fields, Onur Bilgin, **Stephen Steinle**, Yernar Sadybekov, Zaid Marji, John Licato

New Frontiers in Artificial Intelligence, JSAI-isAI 2024, LNCS 14741 (2024), pp. 200-211

Exploring Prompting Approaches in Legal Textual Entailment

Onur Bilgin, Logan Fields, Antonio Laverghetta Jr., Zaid Marji, Animesh Nigohkar, **Stephen Steinle**, John Licato

The Review of Socionetwork Strategies, 18(1) (2024), pp. 75-100

Shot Selection to Determine Legality of Actions in a Rule-Heavy Environment

John Licato, Logan Fields, **Stephen Steinle**, Brayden Hollis

2024 IEEE Conference on Games (CoG) (2024), pp. 1-8

Enforcing Grammar in Code Synthesis with Transformers

Dmytro Vitel, **Stephen Steinle**, John Licato

The International FLAIRS Conference Proceedings, 36 (2023)

Teaching

Introduction to AI Math | Teaching Assistant

University of South Florida | Fall 2026

Honors and Service

Sloan Scholarship | 2025

USF Graduate Fellowship | 2023-2024

United States Navy veteran

Founding member, Society of American Military Engineers, USF Post

Selected Research and Engineering Projects

Curiosity Exploration

Research software | CogSci 2025

Semantic graph analysis for comparing patterns of human exploration in Chain Free Association and Semantic Fluency tasks. The companion repository supports the study of cross-task exploration styles.

[Source code](#)

Persona Heuristic Mapping

Research project | Strategic game evaluation

An evaluation pipeline that translates persona descriptions through psychometric inventories into measurable trait profiles. Tournament protocols and behavior metrics test whether persona conditioning produces consistent shifts across prompts and model variants.

Developer Search and MCP Integration

Amazon Ring | Software Development Internship, 2025

Repository and documentation search combining LLM-expanded keyword retrieval with document summaries. An MCP service enabled Cursor to generate OpenSearch queries and surface keyword-usage summaries across code and internal documentation.

Legal Retrieval and Entailment

Research project | Legal NLP

Retrieval and retrieval-augmented generation pipelines for large case corpora, including segmentation, indexing, query-focused retrieval, and citation-grounded answer synthesis. Evaluation tracks faithfulness, citation coverage, and entailment errors.

Grammar-Constrained Code Generation

Research project | FLAIRS 2023

A constraint layer for transformer code generation that filters token choices using a formal grammar and allowed constructs. Experiments examine constraint strictness, syntactic validity, and exact-match behavior.

Natural Language Action Legality

Air Force Research Laboratory | Machine Learning Internship, 2023

A system that maps natural-language utterances into structured actions and determines whether they are permitted by a ruleset and the current game state.

Technical Skills

Machine learning and NLP: Python, PyTorch, Transformers, Retrieval-augmented generation

Search and data systems: OpenSearch, Indexing, Query construction, Data pipelines

Engineering: Linux, Git, Docker, AWS, Bedrock, IAM

Tools: Cursor, Retool, MCP

Research methods: Psychometrics, Cognitive modeling, Statistical analysis